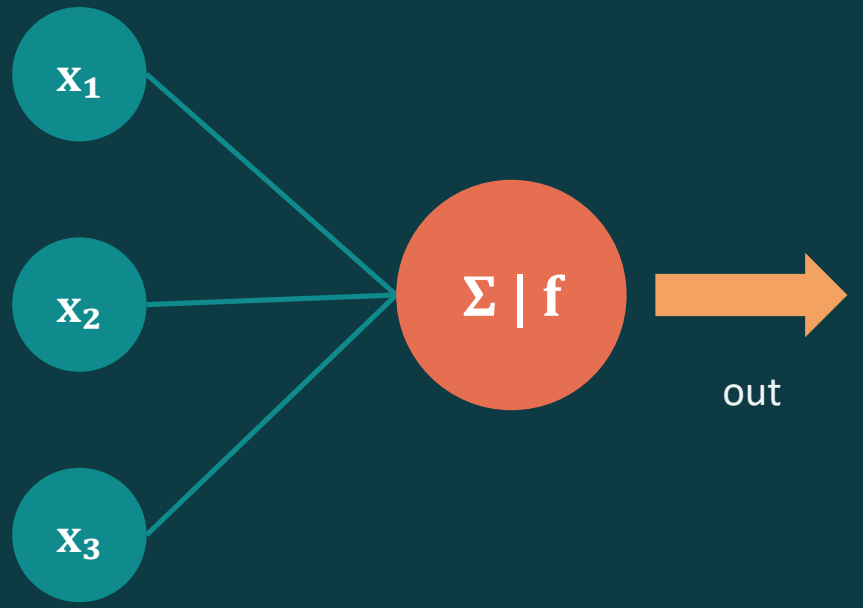
Introduction to Artificial Neural Networks

KnowledgeGATE Bytes

From biological neurons to activation functions



What you will learn today



Biological neuron

Structure of a nerve cell and how it processes signals



Types of neurons

Sensory neurons, motor neurons and interneurons



Artificial neuron

The mathematical model and its five components



Activation functions

Step, signum and sigmoid functions

01

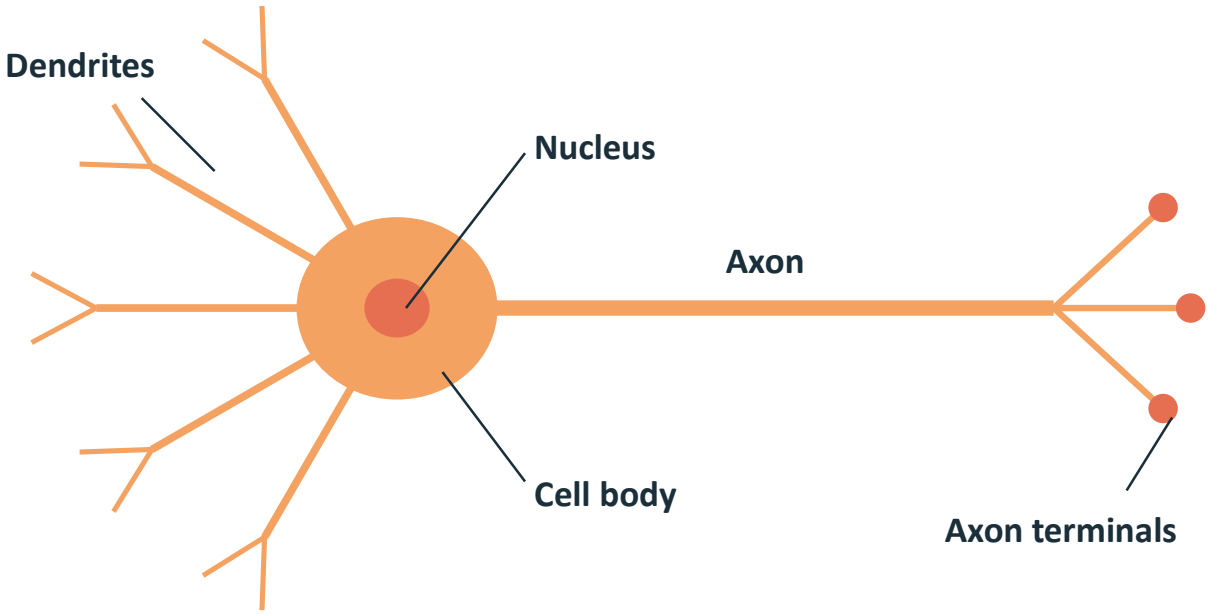
Biological Neurons

The inspiration behind artificial neural networks

What is a neuron?

Neuron, also called a nerve cell

- The basic working unit of the brain
- Specialised cells that send and receive signals from other neurons
- The human brain has roughly 86 billion of them, connected in huge networks



Signals enter through the dendrites, are processed in the cell body and leave along the axon to the axon terminals.

Structure of a biological neuron



INPUT

Dendrite

Receives signals from other neurons.



SUMMING UNIT

Cell body

Sums up all incoming signals and contains the threshold unit.



OUTPUT

Axon

When the sum reaches the threshold, the neuron fires a signal that travels on to other neurons.

Signals flow one way: dendrites → cell body → axon → next neuron

Types of biological neurons



Sensory neurons

Handle the senses: taste, smell, hearing, seeing and feeling.



Interneurons

Intermediate neurons in the spinal cord and brain. Pass signals from sensory neurons to motor neurons.



Motor neurons

Control voluntary and involuntary movements of muscles and organs.

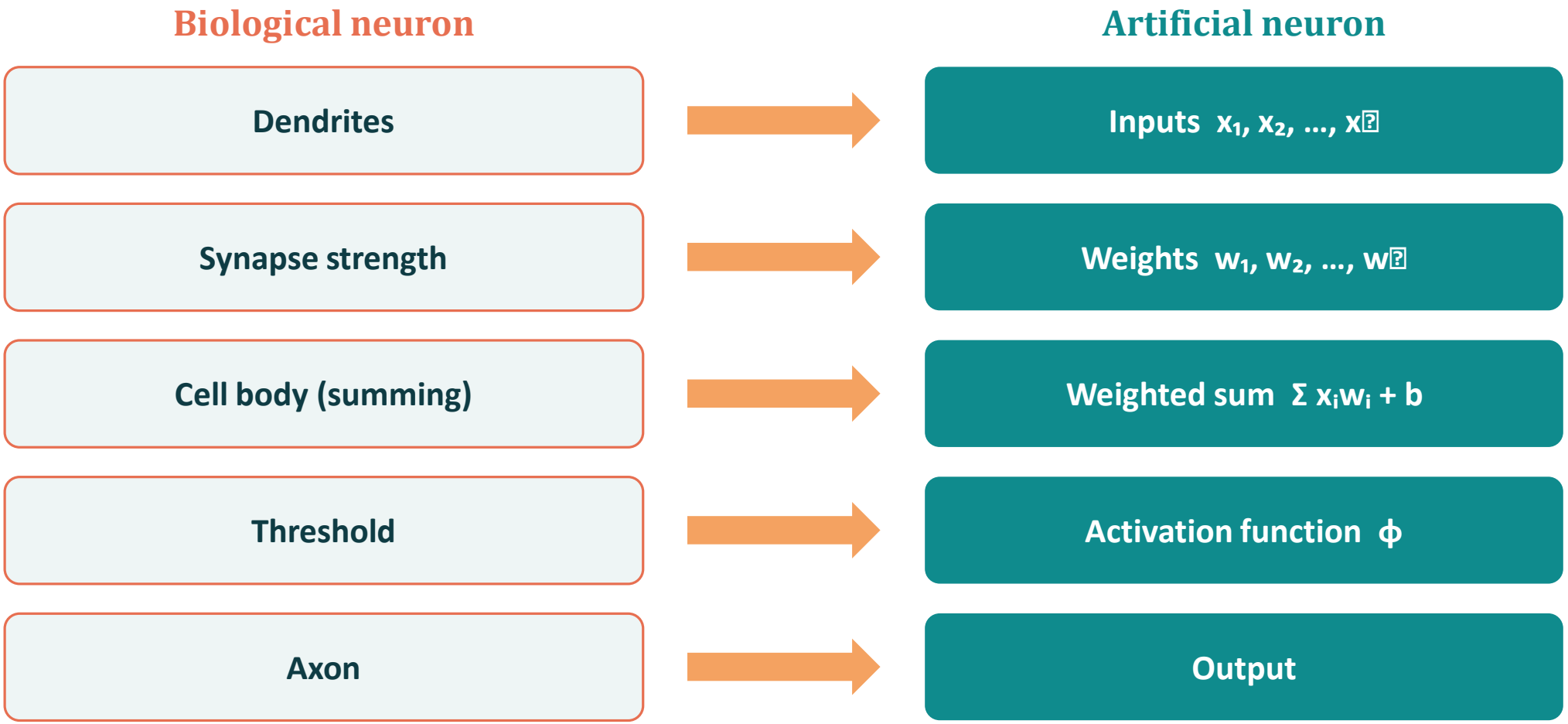
Example: touching a hot surface. Sensory neurons detect the heat, interneurons relay the message, motor neurons pull your hand away.

02

The Artificial Neuron

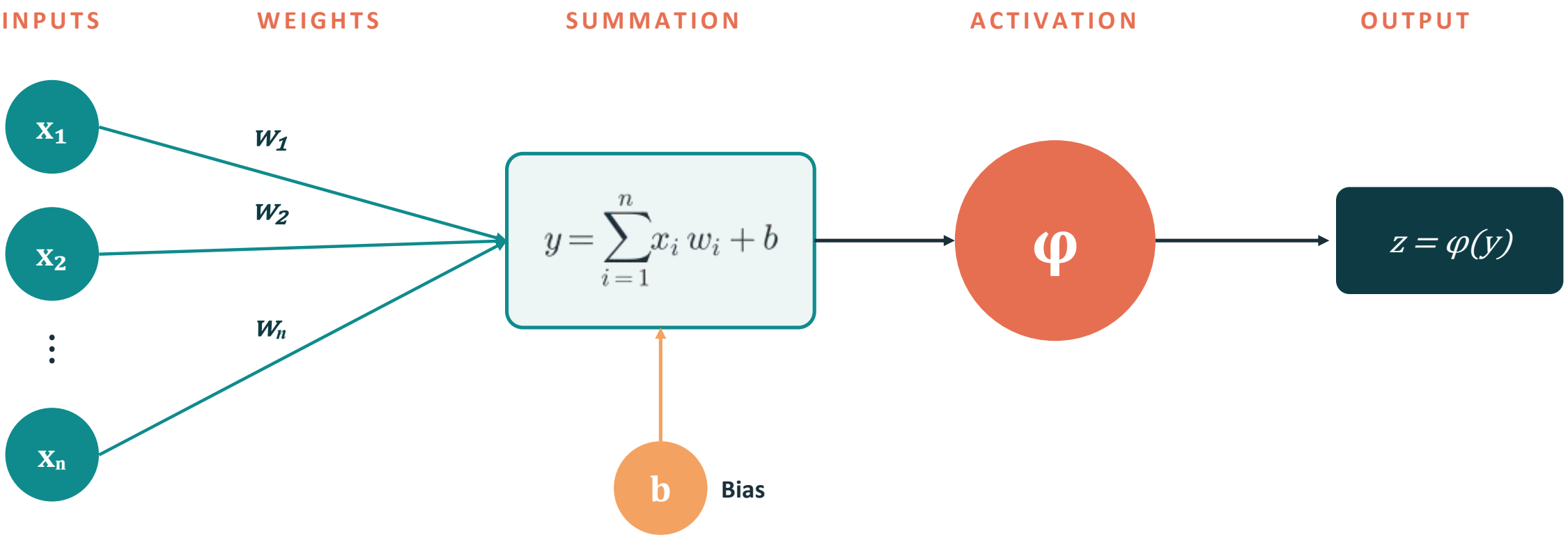
A mathematical model of the biological neuron

From biological to artificial



The artificial neuron model

An artificial neuron is a mathematical function based on the model of biological neurons.



Five components of an artificial neuron

1

Input values

Values passed into the neuron through the input layer. Similar to dendrites.

2

Weighted links

Each link has a weight w_i. Each input x_i is multiplied by its weight to give synaptic strength.

3

Summing function

Adds all weighted inputs: $\Sigma\, x_i w_i$. A bias b is also added to this sum.

4

Activation function

Decides whether the neuron fires, based on the summed value.

5

Output

The final output of the neuron, which can be passed on to other neurons.

Signal path: inputs → weights → sum + bias → activation → output

Worked example: one neuron, step by step

Given

Inputs

$x_1 = 1,\ x_2 = 0.5$

Weights

$w_1 = 0.6,\ w_2 = -0.4$

Bias

$b = 0.1$

1 Weighted sum

$$y = (1)(0.6) + (0.5)(-0.4) + 0.1 = 0.5$$

2 Apply an activation function

Step

$y > 0$, so $z = 1$

Neuron fires

Sigmoid

$z = \sigma(0.5) \approx 0.62$

62% activation

03

Activation Functions

Deciding whether a neuron fires

What is an activation function?

Activation functions are mathematical functions that determine the output of a neuron. They decide whether a neuron should be activated ("fired") or not.



Decides firing

Converts the weighted sum y into the neuron's output z.



Adds non-linearity

Without it, a network of any depth behaves like one linear equation.



Handles complex data

Non-linearity lets networks learn curved, complex patterns.

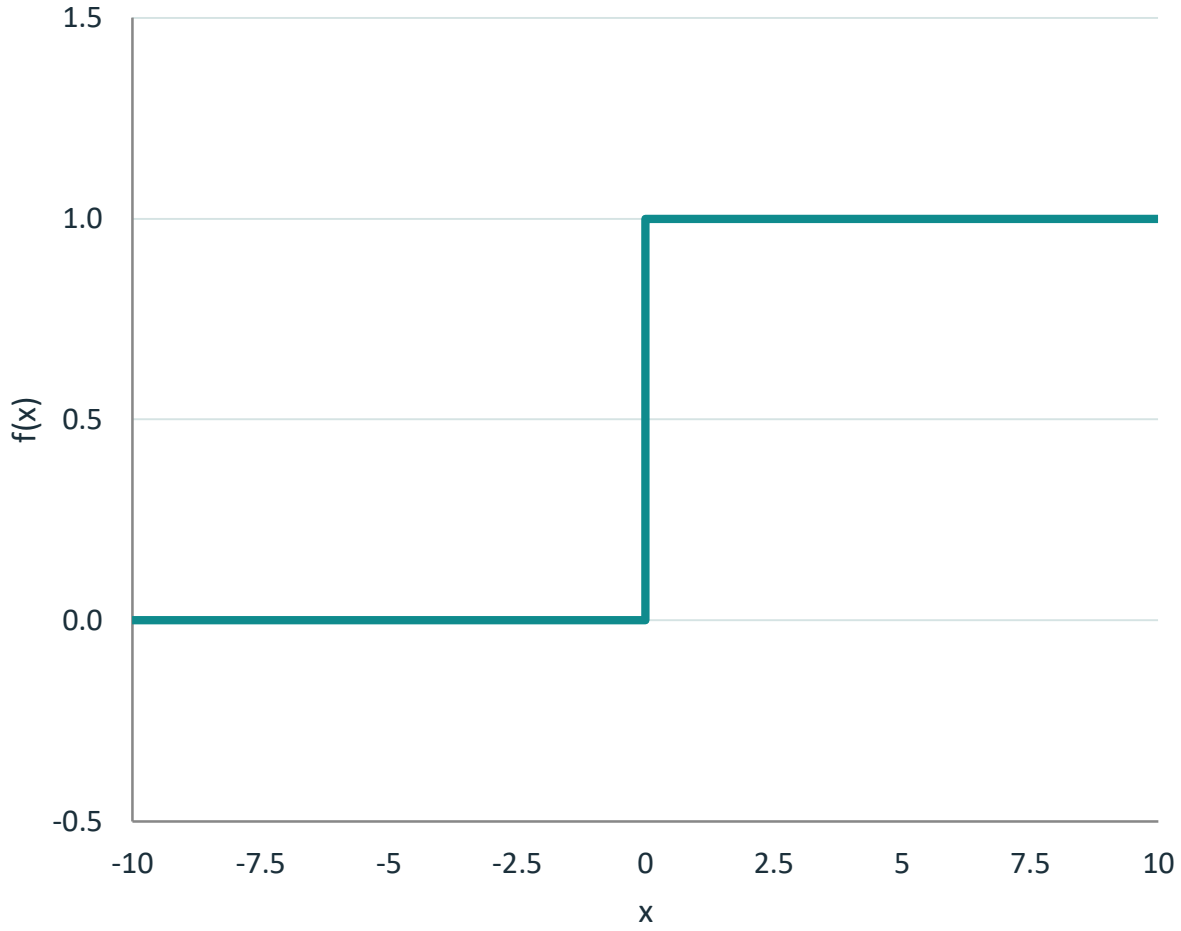
Step function

$$z = f(x) = \begin{cases} 0, & x \leq 0 \\ 1, & x > 0 \end{cases}$$

Output 0 or 1

Type Hard threshold, binary

Used in The classic perceptron



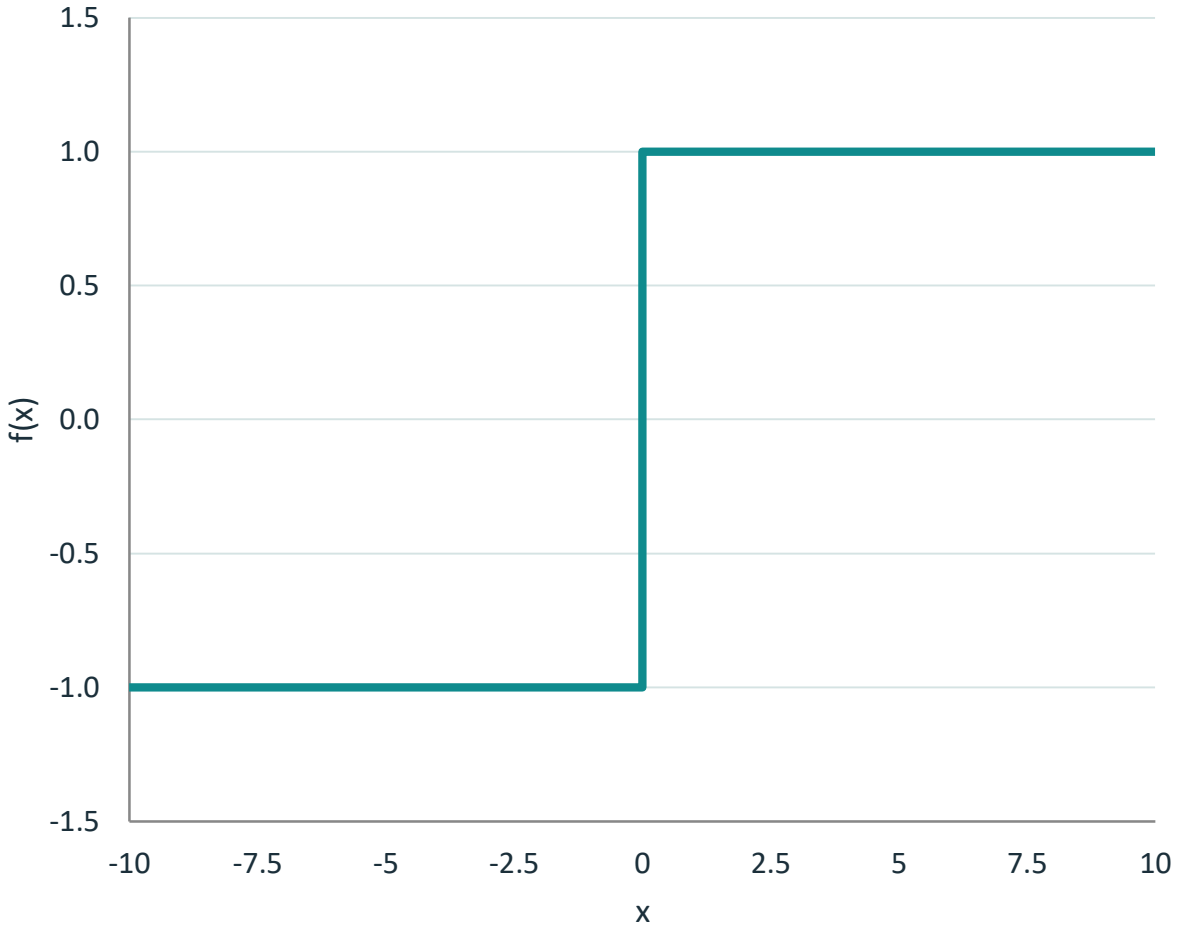
Signum function

$$z = f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$$

Output −1, 0 or +1

Type Hard threshold, bipolar

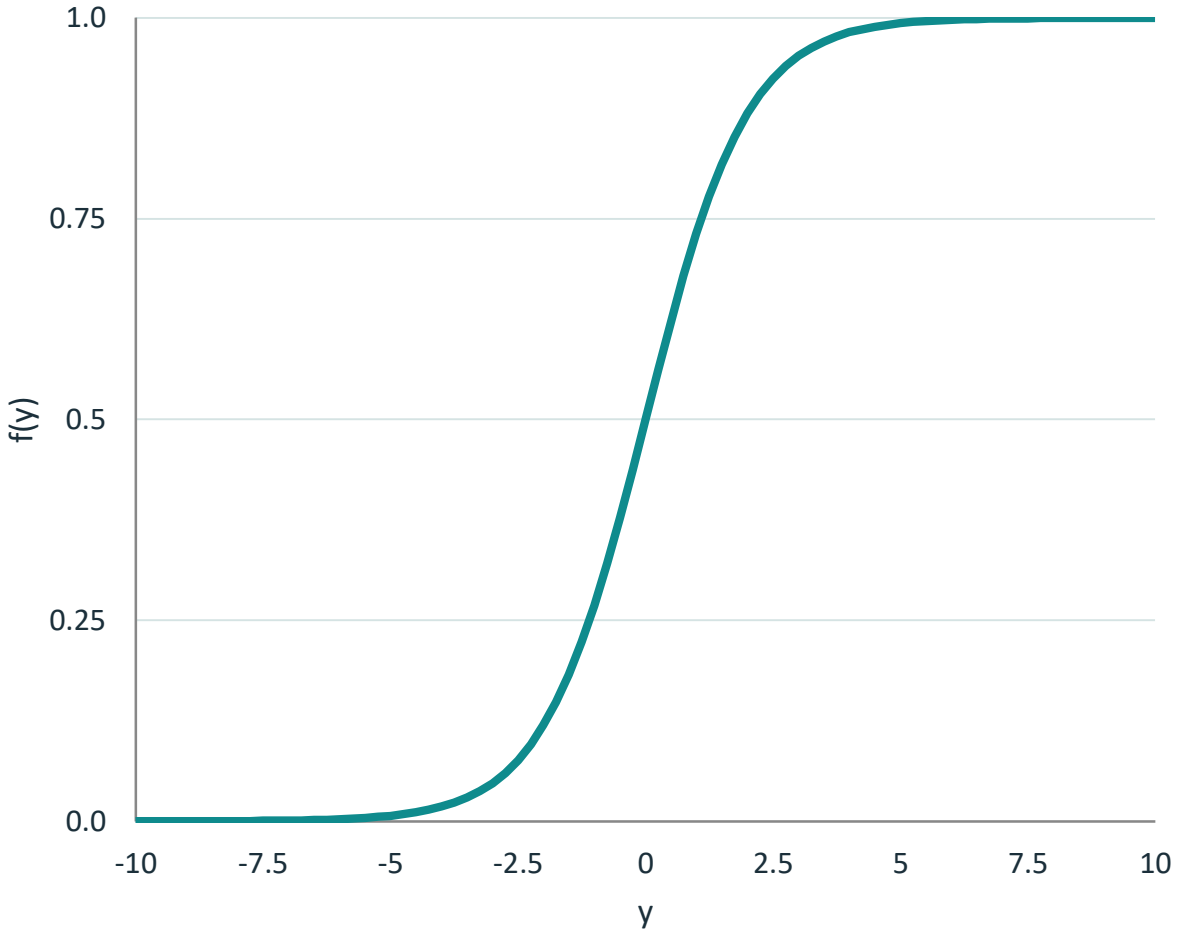
Used in Bipolar outputs (−1 / +1)



Sigmoid (logistic) function

$$z = f(y) = \frac{1}{1 + e^{-y}}$$

Output	Between 0 and 1
Type	Smooth S-curve, differentiable
Used in	Binary classification output (probability)



Comparing the three functions

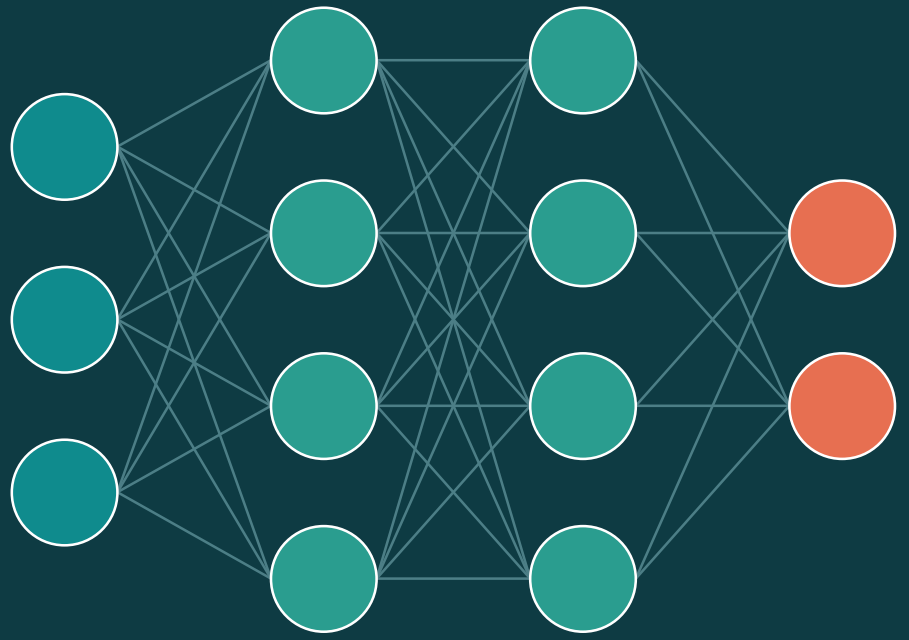
Function	Output values	Smooth?	Typical use
Step	{0, 1}	No	Perceptron, binary on/off decisions
Signum	{−1, 0, 1}	No	Bipolar outputs centred on zero
Sigmoid	(0, 1)	Yes	Binary classification probabilities



Key idea: Step and signum give hard yes/no answers. Sigmoid gives a smooth, graded answer, which is what makes training by backpropagation possible.

Artificial Neural Networks

Definition, architecture and types of ANN



What you will learn today

What an ANN is

Definition and key features

How it learns

A worked example and learning from data

Architecture

Input, hidden and output layers

Types of ANN

Single-layer, multilayer and recurrent networks

EXPLANATION

In the last class we modelled a single artificial neuron. Today we connect many of these neurons together to form a network, and see how such a network is organised and what kinds of networks exist.

01

What is an ANN?

Definition, a worked example and learning from data

Artificial Neural Network (ANN)

Brain-inspired

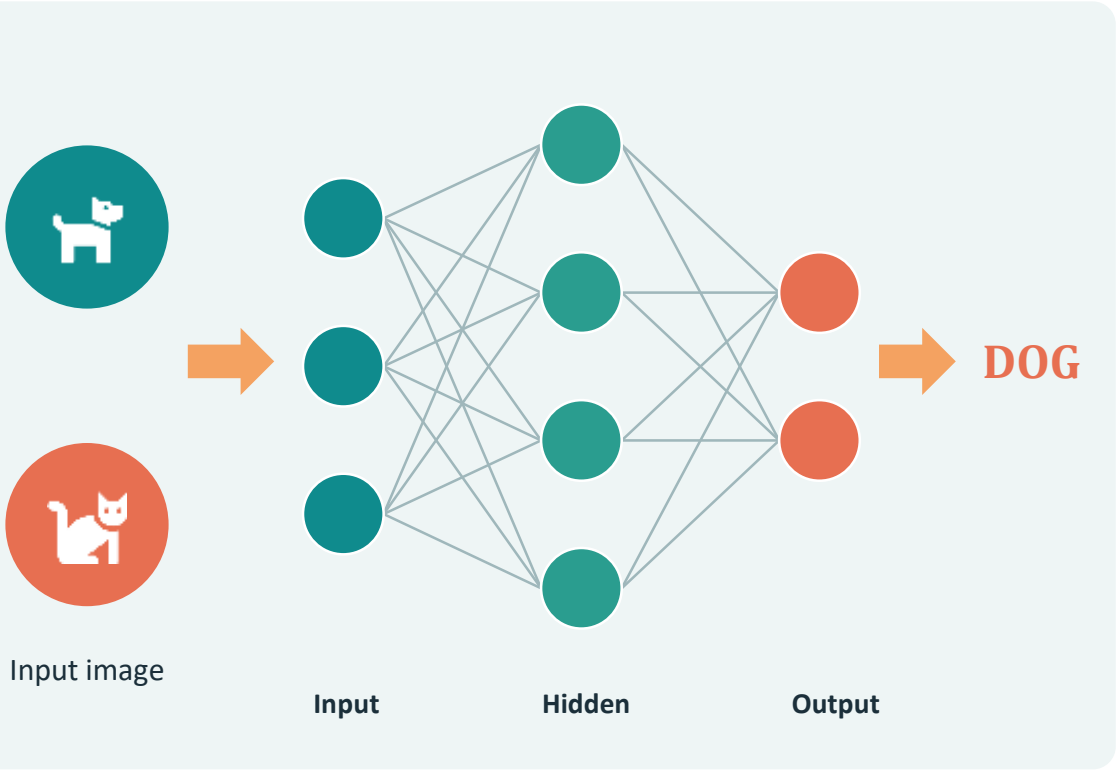
A computing system designed to simulate the way the human brain analyses and processes information.

Connected neurons

Made of many interconnected processing elements called artificial neurons (nodes), joined by connection links.

Self-learning

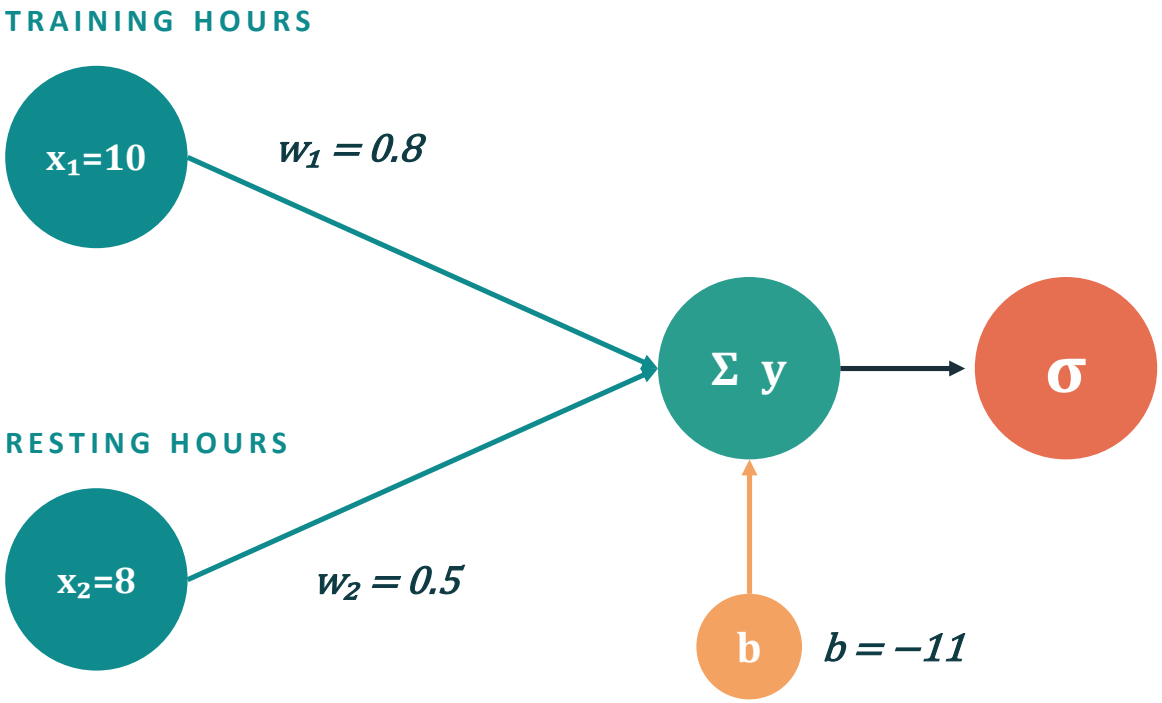
Improves its performance as more data becomes available.



EXPLANATION

Show the network thousands of labelled cat and dog pictures. Each time it guesses wrong, it adjusts its connection weights a little. After enough examples it can correctly label a new picture it has never seen before.

Example: how a neuron in the network computes



1 Weighted sum

$$y = x_1 w_1 + x_2 w_2 + b$$
$$= 10(0.8) + 8(0.5) - 11 = 1$$

2 Sigmoid activation

$$z = \sigma(y) = 1 / (1 + e^{-y})$$
$$= 1 / (1 + e^{-1}) \approx 0.73$$

Output 0.73 → about a 73% chance of a good result

EXPLANATION

Each input (training hours, resting hours) is multiplied by a weight showing how important it is, then added with the bias. The sigmoid squeezes the result into 0 to 1, so the output can be read as a probability.

Where do the weights come from?

The network learns them from data

↻ repeat

EXAMPLE DATASET: 10,000 RECORDS



Start

Random w_1, w_2, b

Predict

Compute output z

Compare

Check against the true answer

Adjust

Change weights to reduce error

EXPLANATION

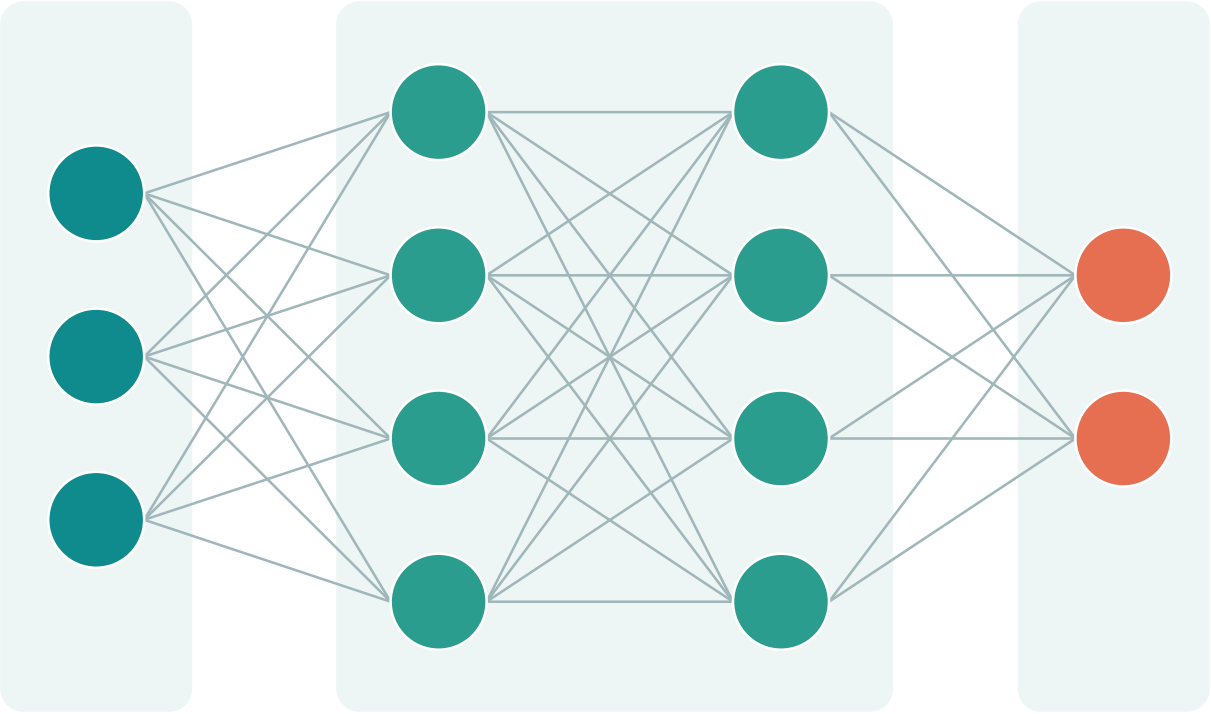
We never choose the weights by hand. The network starts with random weights and repeats predict → compare → adjust on the 8,000 training records. The 2,000 test records are kept aside to check how well it works on data it has never seen.

02

Architecture of ANN

Input, hidden and output layers

Architecture of an ANN



Input layer

Nodes that receive input from the outside world.

Hidden layers

Lie between input and output; do the intermediate computations.

Output layer

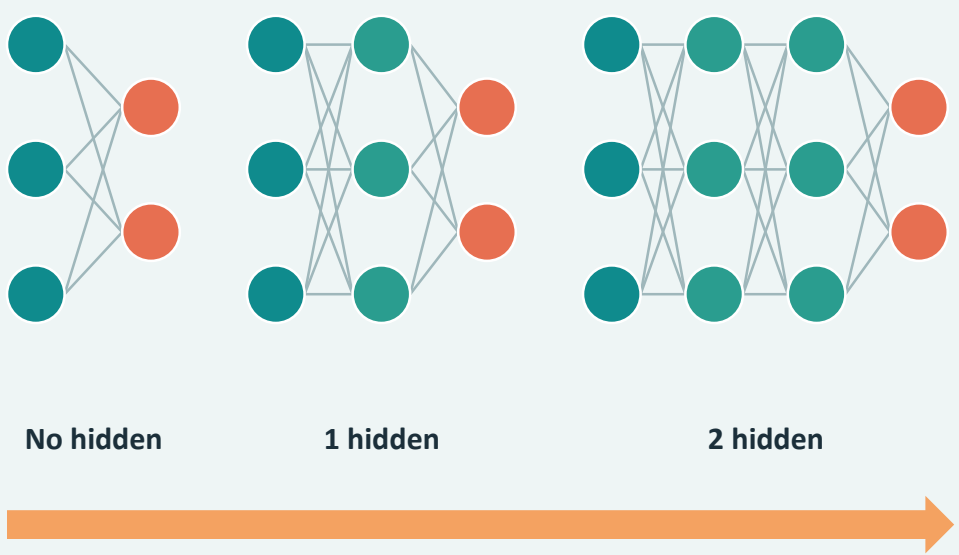
Nodes that provide the output of the network.

EXPLANATION

Data enters at the input layer, is transformed step by step in the hidden layers, and the answer comes out of the output layer. Each node is connected to every node in the next layer, and every connection has its own weight.

Hidden layers

1. Located between the input and output layers
2. Made of hidden nodes that are not directly visible to the external system
3. Responsible for intermediate computations and feature extraction
4. Zero or more hidden layers, depending on the complexity of the problem
5. One hidden layer is enough for many problems; deeper networks model more complex patterns



EXPLANATION

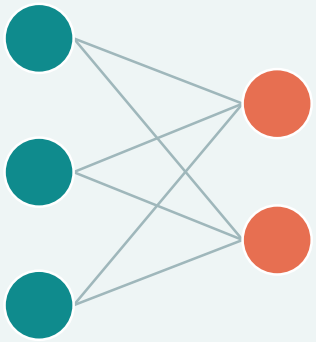
In a face-recognition network, the first hidden layer might detect edges, the next combines edges into eyes and noses, and the output layer decides whose face it is. Each hidden layer builds new features from the ones found before it.

03

Classification of ANN

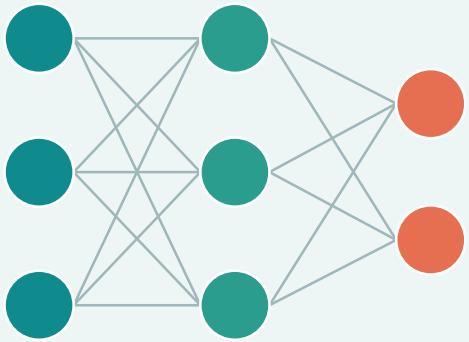
Single-layer, multilayer and recurrent networks

Classification of artificial neural networks



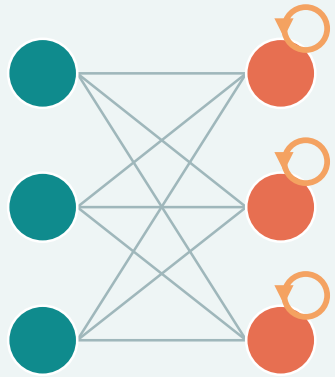
Single-layer feed-forward

One computational layer; information flows forward only.



Multilayer feed-forward

One or more hidden layers; information flows forward only.



Recurrent network

Has at least one feedback loop.

EXPLANATION

Two questions classify a network: how many computational layers does it have (one, or several), and does information only move forward, or can it loop back? Feed-forward networks have no loops; recurrent networks do.

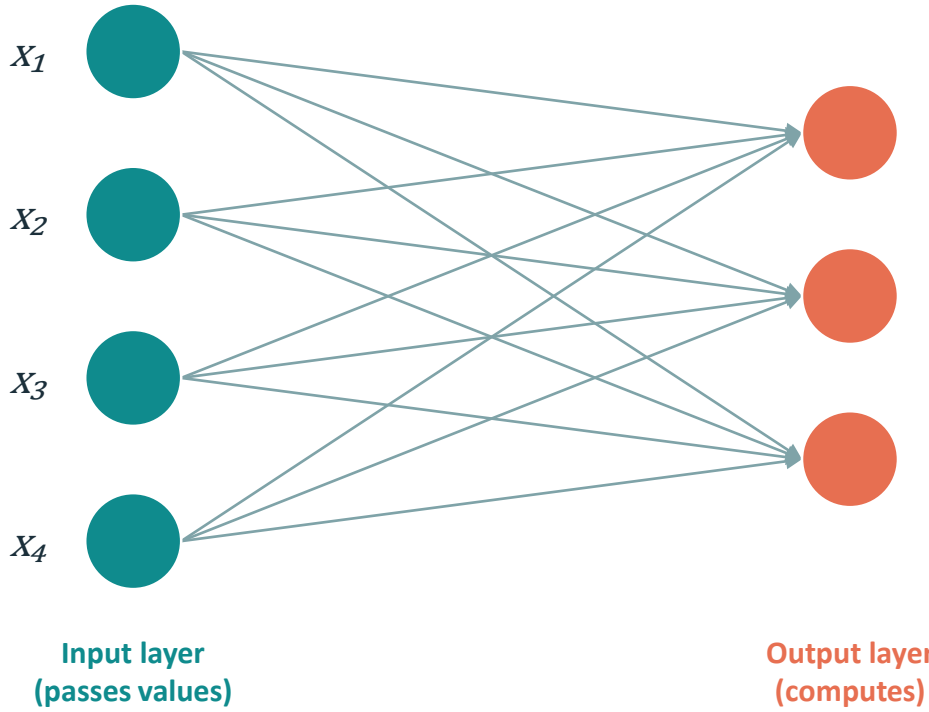
Single-layer feed-forward network

Single layer

Only one computational layer (the output layer), so it is called single-layer, even though it has two layers in total: input and output.

Feed-forward

Information flows from the input layer to the output layer without any feedback loops.



EXPLANATION

We don't count the input layer because it only passes values in without calculating anything; only the output layer computes. The perceptron is the classic example. Single-layer networks can only solve simple, linearly separable problems.

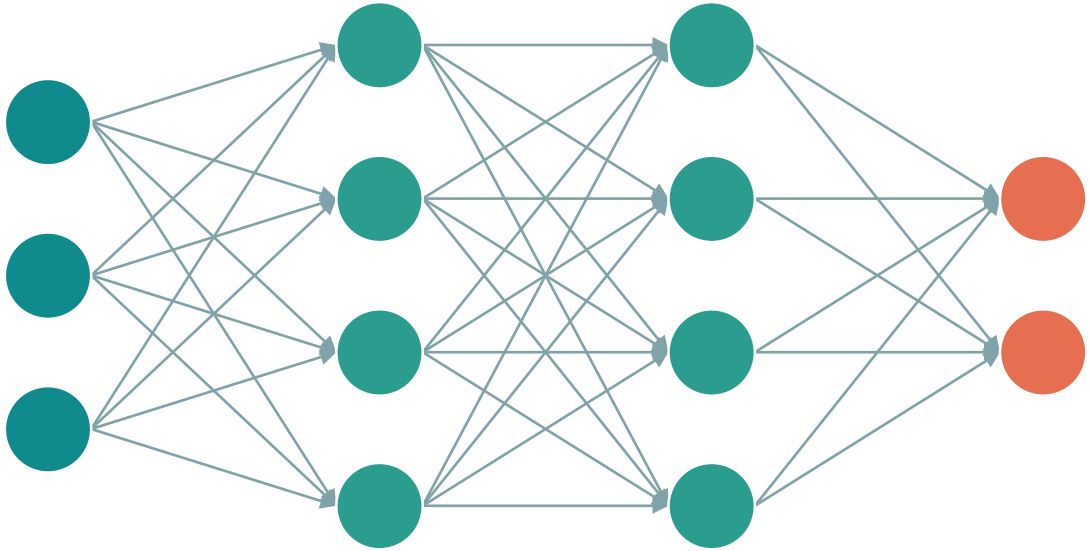
Multilayer feed-forward network

Multilayer

Includes an input layer, one or more intermediate (hidden) layers, and an output layer.

Hidden computation

Hidden layers perform intermediate computations before passing the information to the output layer.



EXPLANATION

Hidden layers let the network learn complex, non-linear patterns, such as XOR, that a single layer cannot. Information still moves strictly forward; nothing loops back. The Multi-Layer Perceptron (MLP) is this type.

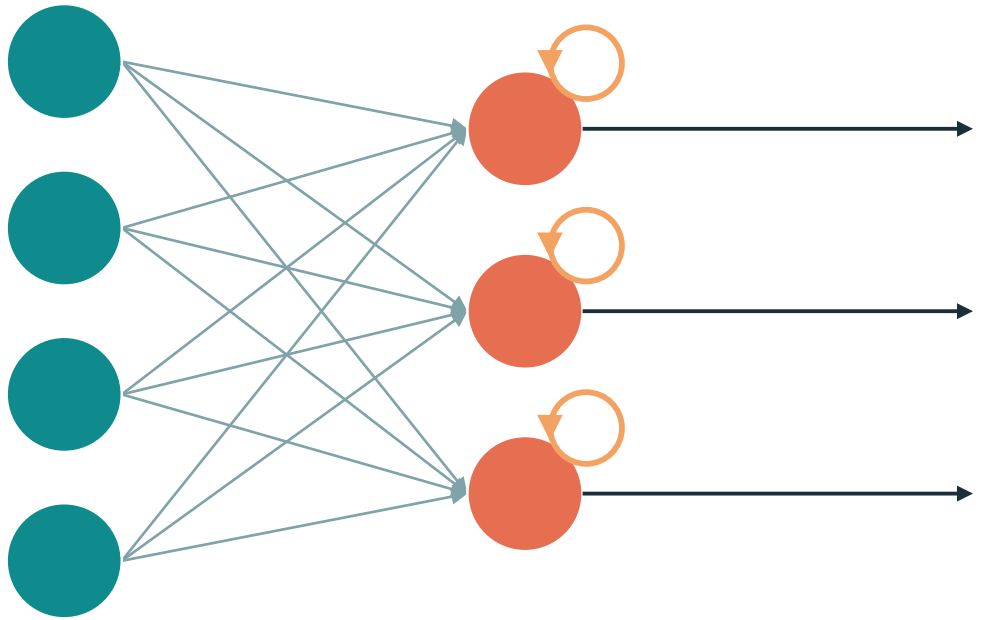
Recurrent network

Feedback loop

Includes at least one feedback loop, which makes it different from feed-forward architectures.

Single or multilayer

Can be built as a single-layer or a multilayer recurrent network.



Output fed back as input at the next step

EXPLANATION

The feedback loop gives the network a kind of memory: its output depends on the current input and on what it saw before. That suits sequences such as text (predicting the next word), speech, or prices over time.

Comparing the three types

Type	Layers	Information flow	Memory	Good for
Single-layer feed-forward	Input + output (1 computes)	Forward only	No	Simple, linearly separable problems
Multilayer feed-forward	Input + hidden + output	Forward only	No	Complex patterns, e.g. image classification
Recurrent	Single or multilayer	Forward + feedback loops	Yes	Sequences: text, speech, time series



EXPLANATION

Feed-forward networks (single-layer and multilayer) treat each input independently, so they suit tasks like classifying a single image. Recurrent networks remember earlier inputs, so they suit data where order matters.

Key takeaways

Artificial Neural Networks

1. An ANN is a brain-inspired network of connected artificial neurons that learns from data
2. Each neuron computes $y = \Sigma x_i w_i + b$ and applies an activation such as sigmoid
3. Weights are learned: predict, compare, adjust, repeat; then test on unseen data
4. Layers: input (receives), hidden (computes, extracts features), output (answers)
5. Types: single-layer and multilayer feed-forward (no loops), and recurrent (feedback loops)